Gamma, Delta and, now, Omicron. But whether they get a Greek letter or not, the Omicron family has been creating waves. "Its subvariants have been causing surges in other countries, so one gets to hear of them more," says infectious disease expert Dr Kirti Sabnis.

Some argue that Omicron's subvariants deserve individual Greek letters. Sub-variants such as BA.4 and BA.2.12.1 can overcome immunity bestowed by other Omicron sub-variants. This will also help the general public understand and heed pandemic control measures. However, the WHO has said these sub-variants aren't different enough in their mutations to be dubbed entirely new variants. Delta, too, had its share of sub-variants. The one that found most mention was AY.4.2, a.ka. Delta Plus. However, while just one variant of the Delta family triggered a single surge in the world, different variants of Omicron are causing different surges in countries.

MUTANT THREAT

A similar contest took place when Covid first arrived, with various variants fuelling surges in different parts of the world. India was afflicted with the original strain from Wuhan. However, South Africa had Beta and Brazil Gamma triggering their first waves. Omicron's sub-branches now seem to be doing the same. More than 100 subvariants have been found till date, each trying its best to overpower the other by mutating and succeed in replication.

BA.2 has become the most powerful Omicron variant, dominating worldwide with 21 sub-variants of its own. It gained this status due to the wealth of mutations within and external to its spike protein. The BA.2 genome contains 53 amino acid mutations, 29 of which are in the spike, far greater than mutations in previous variants. These mutations allow it to be more contagious, and evade antibodies in humans from previous infections and/ or vaccines. However, BA.2 now seems to have a new challenger. In the US, BA.2.12.1 seems to be taking over from BA.2 like the latter did from BA.1. A recent analysis by the Centers of Disease Control and Prevention in the US suggests that BA.2.12.1 accounts for at least 19 per cent of the new cases while BA.2 lineages overall make up 90 per cent of the cases. The country has also had a BA.1-triggered wave earlier.

If the first Omicron variant was three times more effective in reproducing than Delta, then BA.2 is 30 per cent more replicative than the first and BA.2.12.1 25 per cent more than its parent. The sub-variants are also trying to infect host cells beyond the nasal tissue. BA.2.12.1, BA.4 and BA.5 have an L452 mutation, which is of particular concern to experts, as this allows the virus to infect the nasal and lung tissues, unlike the original Omicron that impacted only the nasal tract.

Interestingly, Omicron has also sparked a number of hybrid or recombinant variants that are a mix of two sub-variants. The cross between BA.1 and BA.2 has produced XE, XG, XH, XJ, XK, XL, XN, XP, XQ and XR. Omicron and Delta have formed the recombinants XD and XF. While these have been spotted across the world, they aren't drastically different from the parent variant in the case of the Omicron recombinant. But the Delta-Omicron hybrid contains structural proteins from Delta and experts in France and the UK where these two have been found are monitoring them closely. "We don't know which variant will have enough new mutations to become an entirely new variant. Even Omicron's sub-variants have the potential to harm high-risk groups—the elderly, those with co-morbidities, and those unvaccinated," says Dr Farah Ingale, internal medicine specialist at the Fortis hospital in Vashi, Navi Mumbai.



OMICRON'S **SUB-VARIANTS HAVE BEEN CAUSING SURGES IN OTHER COUNTRIES**, SO ONE GETS TO HEAR OF THEM MORE

DR KIRTI SABNIS
Infectious disease consultant, Fortis, Vashi



THERE'S NOINDICATION THAT THE CURRENT LOT OF SUB-VARIANTS IS UNUSUAL OR WILL CAUSE SURGES. BUT IT SHOWS **COVID'S TRYING TO SURVIVE**. THERE COULD STILL BE A NEW VARIANT

DR RAKESH MISHRA
Director, TIGR, Bengaluru

SHOULD INDIA WORRY?

Currently, both natural and vaccine-acquired immunity offer some protection against severe disease and death, feeding the perception of Omicron being mild in India. The experience in Hong Kong, though, was different. With the arrival of BA.2.2, it recorded 1.1 million infections and over 8,000 deaths compared to 12,631 confirmed